PHOTOELECTRIC DUAL BEAM DETECTOR ABT-i100

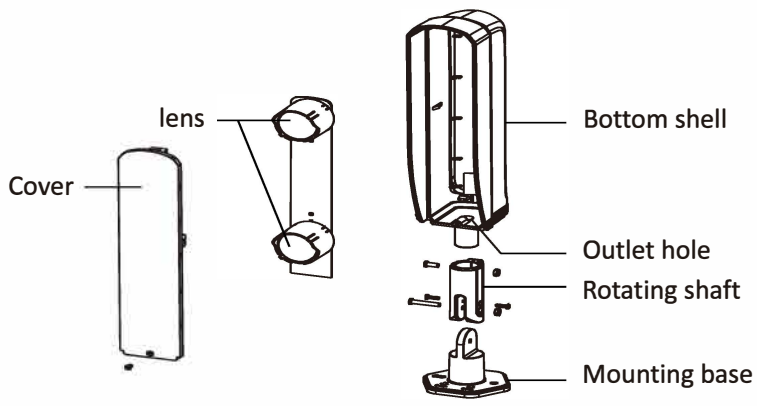
◆ Thanks for purchasing photoelectric IR beam detector, please read the user manual carefully before installation.

Türkçe Kullanım Klavuzunu Görüntülemek için , Lütfen QR kodunu cep telefonunuza okutunuz

1.Features

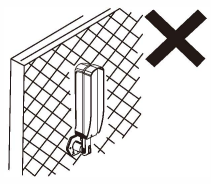
- Interruption time or walkspeed adjustable
- Self-contained mounting bracket for quick and easy installation
- Frequencies selectable for long distance and stacking installations
- Wide voltage power input: DC9-30V/AC9-20V
- Waterproof grade: IP65
- Alignment angle horizontally (±360°), vertically (±180°)
- Digital filtering, high environment adaptability to eliminate false alarms
- NO / NC relay outputs
- Specific mode indicator for simple and accurate optical correction
- The design of the brim extending outward makes the anti-sunlight interference ability stronger

2.Part Description

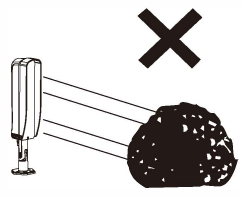


3.Installation Notes

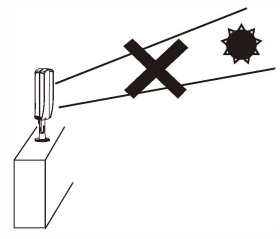
(1).Please avoid below situations to assure performance



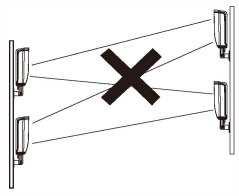
1. Do not install on an unsteady or moveable base



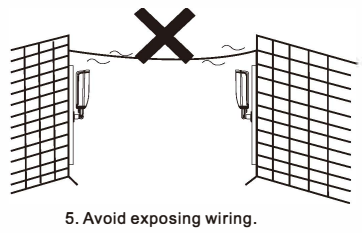
2. Do not install the unit where objects can block the beams like plants and laundry moving in the wind.



3. Prevent direct sunlight onto the receiver



4. Avoid cross talk. Use frequency select (stack installation only for the same model)



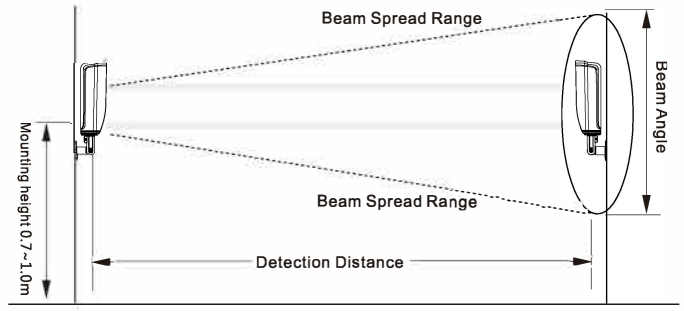
5. Avoid exposing wiring.

(2).Normal installation

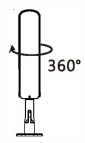
◆ Detection distance

Model	Detection Distance	Beam Angle
Single Beam	20m	0.8m
	40m	1.2m
Dual Beams	60m	1.2m
	100m	1.8m
	100m	1.8m

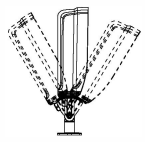
◆ Mounting height



◆ Adjusting angle



Horizontal 360° (±180°)

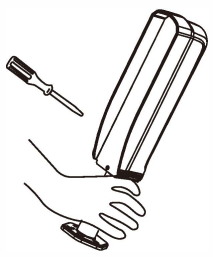


Vertical 180° (±90°)

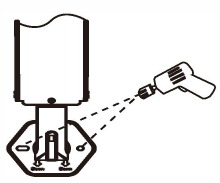
Notice: For best testing results, please avoid testing in 45°

4.Setting Method

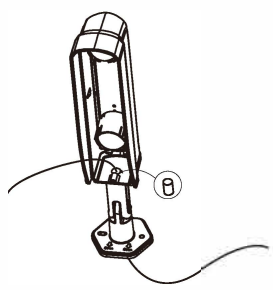
◆ Wall mounting



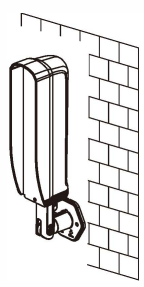
1. Loosen the screw and remove the cover



2.According to the size of the base mounting hole, make two mounting holes on the wall.



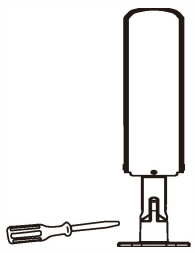
3.Threading: the reserved embedding thread is taken out from the base upwards, and the appropriate length is 10cm to prepare the wiring, and then the sponge plug is inserted into the hole.



4. Attach beam to the base



5. Connecting wires to the terminals (please refer to "beam alignment")

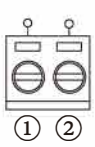


6. Review and reset the cover

5.Connectors

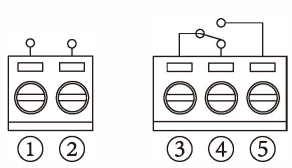
Warning Do not exceed the voltage or current rating specified for any of the terminals during installation.

Transmitter:



POWER
DC9-30V/AC9-20V

1 . Power input: DC9-30V/AC9-20V.

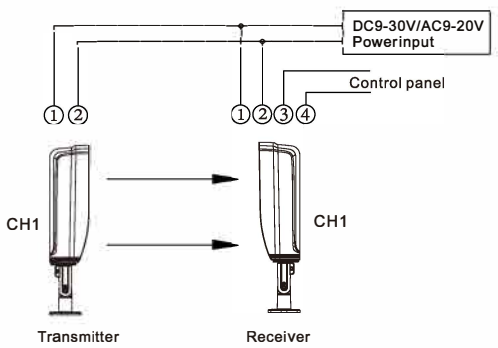


POWER
DC9-30V/AC9-20V
COM NC NO
ALARM

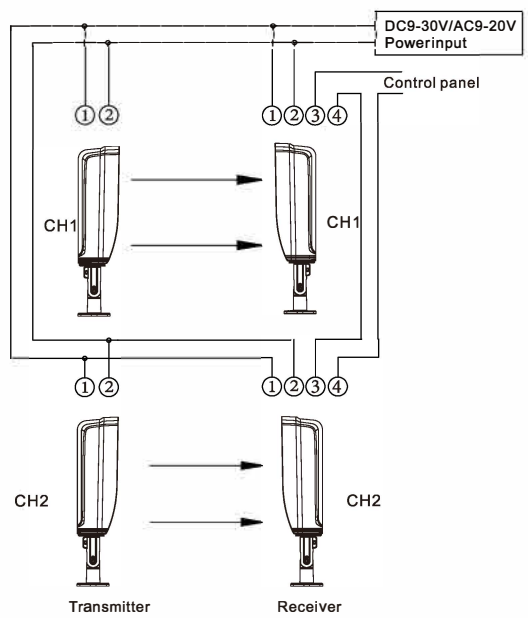
1 . Power input: DC9-30V/AC9-20V.
2. C relay (30VDC 1.0Amax).

6.Connecting Wires

(1).Single connect: Control panel operating voltage DC12V, NC alarm output. Connecting to power supply parallel

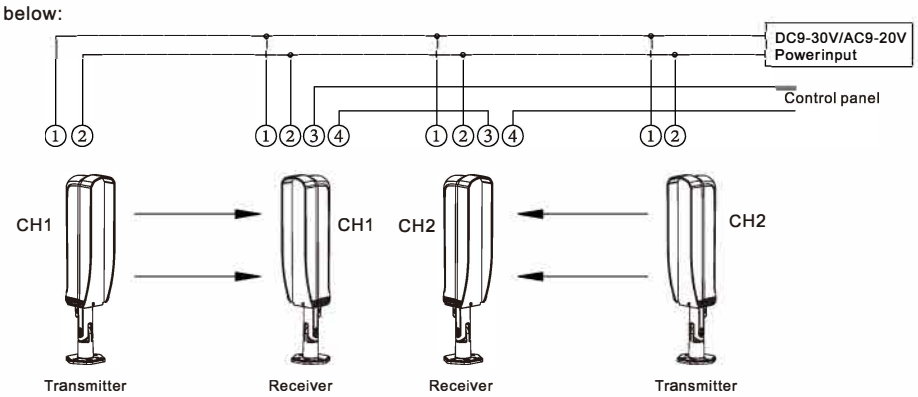


(2).Stacked connect: Control panel operating voltage DC12V,NC alarm output series connect



(3).2 pairs install in series: Connect power of transmitter and receiver in series with 12V DC on power supply. Alarm output is N.C.

As below:



◆ Wiring distance between the power supply and the detector should not exceed the following table length.

Wire diameter \ Voltage / Length	DC12V
0.5mm² (Φ0.8)	400m
0.75mm² (Φ1.0)	600m
1.0mm² (Φ1.2)	800m
1.5mm² (Φ1.4)	1000m

Warning

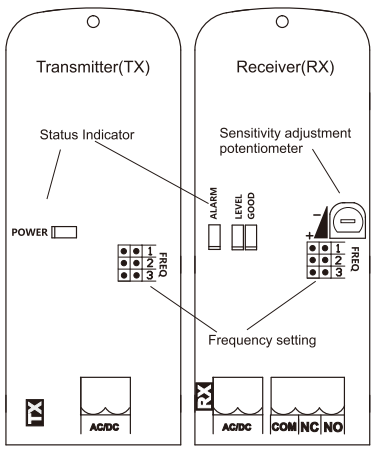
1.The power wire can't exceed the listed length.

2.When connecting multiple detectors, the required cable length is divided by the corresponding number of units listed.

3.Don't connect the port with the voltage or current which is over the normal specification.

7.DIP Switch Explanations

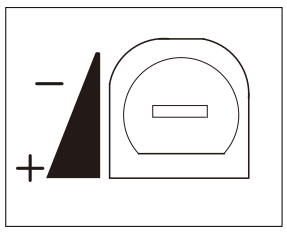
DIP switch show on the left side of the main PCB, as shown in following figure.



1.Frequency setting:
The frequency is 1, 2, 3 bands respectively. When setting the beam frequency, the emitter, the receiver jumper cap must be inserted in the same frequency band.(When two pairs or more pairs of pairs are mounted on the same line or on the same surface, it is recommended that the adjacent pairs of pairs are set to different frequencies to prevent mutual crosstalk.)

2. Indicator status
Signal strength status:
When there is no signal, the indicator does not light; the blue (LEVEL) indicator is set to off, slow flash, fast flash, and always bright. When the RX receives the infrared signal from scratch, the signal strength changes from weak to strong, and the blue (LEVEL) indicator indicates the infrared signal strength in the manner of the four levels of brightness.
The green (GOOD) indicator is set to four states: off, slow flashing, fast flashing, and always on. When the infrared signal reaches a certain intensity, the blue (LEVEL) indicator goes off, and the green (GOOD) indicator enters a slow flashing state, and the signal strength is stronger. The green (GOOD) indicator flashes faster until the signal strength is optimal and the green (GOOD) indicator is always on.
Power indicator (POWER): After power-on, the green indicator lights up.
(Note: After about 30 minutes, the indicator goes out, it does not affect the TX's transmitting function. After the power is turned back on, the indicator is turned on again)
Alarm indicator (ALARM): When the alarm is activated, the red indicator is on;
When alarm is disactivated, it is off.
(Note: When the alarm is normal, after about 30 minutes, the green indicator goes out, it does not affect the receiver receiving function. After the alarm is occured, the indicator is turned on again.

3. Sensitivity adjustment:
Sensitivity is how long the infrared beam is blocked, which will trigger an alarm. Rotary potentiometer, the interruption time can be adjusted within the range of 40-1000ms.
This feature allows you to adjust the sensitivity of your device to suit your field environment. The beam cut-off time is adjusted based on the speed at which the detector detects the object.
For cases where birds or newspapers may accidentally cut off the infrared beam, a longer cut-off time can be set;
After adjusting the cut-off time, please verify.



The sensitivity adjustment potentiometer sets the sensitivity level by adjusting the shading time:

- Clockwise adjustment, masking time is reduced, sensitivity is increased;
- Counterclockwise adjustment increases the masking time and reduces sensitivity.

10.Troubleshooting

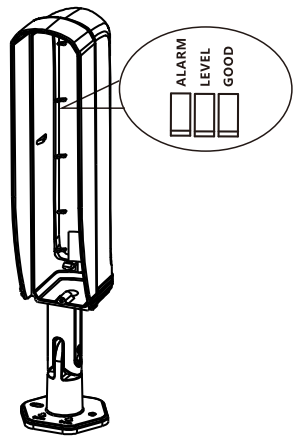
Symptom	Possible cause	Remedy
Power on, but power LED off	1. No voltage on power cable; 2. Broken circuit or short circuit; 3. Beyond specified voltage; 4. Power cable exceeds the specified length	1. Check PSU, voltage, cables and connectors
When beam is blocked, the alarm LED does not indicate, nor does the alarm relay switch	1. There is reflection or cross-talk from other transmitters 2. Walk speed set too long 3. Alarm output cable is shorted or damaged	1. Change beam path or change TX/RX frequency channel 2. Ensure 2 beams all blocked 3. Change walk-speed setting 4. Check RX terminal and output cable
When beam is not blocked, alarm LED indicates activation	1. Beam is out of alignment; optical axis does not overlap 2. There are objects between TX and RX 3. Frequency is incorrect 4. The cover is dirty or capped by snow, frost and ice 5. TX is faulty or OFF	1. Adjust optical axis 2. Check objects between TX and RX 3. Ensure the frequency of TX and RX is the same 4. Clean cover or user heater 5. Check the voltage or wiring of TX
False alarm	1. Bad wiring and fluctuant power voltage 2. Randomly blocked, like birds, paper or leaves 3. The beams base is unstable 4. Out of alignment	1. Check power, current and wiring 2. Change installation location 3. Strengthen installation base 4. Re-align

8.Optical Axis Correction

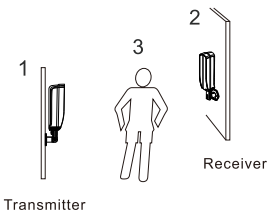
1. Set the infrared beam frequency, the TX frequency must be consistent, such as the TX frequency jumper cap inserted in the band 1, the RX's jumper cap must also be inserted in the band 1.

2. Adjust the shaft so that the TX and RX are aligned as much as possible.

3. Rotate the TX and RX shafts to correct the direction of the infrared beam. When the signal intensity blue **(LEVEL)** indicator changes from slow flash to fast flash, it indicates that the optical axis alignment accuracy is higher. Until the light is normal on. Continue to adjust, the blue indicator is off, and the green **(GOOD)** indicator changes from slow flash to fast flash until the green indicator is on, which means the signal strength is optimal



9.Walk Test

 Alarm Status	Please make sure that the alarm indicator is off. If the infrared beam is not blocked, the red alarm indicator will still light up. Please perform optical correction again.
	Please do the test at the 3 following positions 1. At the transmitter side 2. At the receiver side 3. In the middle
 Alarm Status	Alarm indicator turns on when beam is interrupted, test finished successfully.

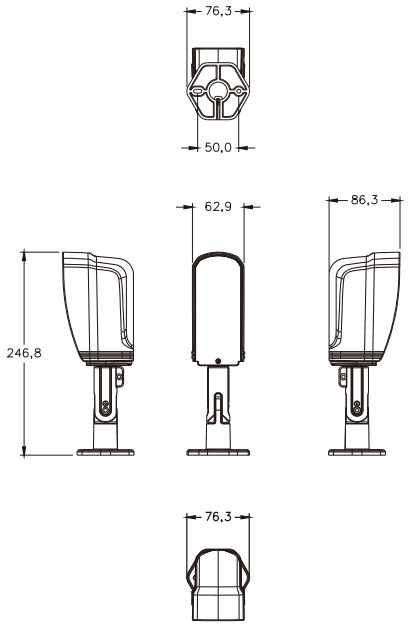
Note: If the alarm LED indicator is OFF even though the beams are completely blocked, refer to the “Trouble Shooting”.

11.Specifications

Model		Single Beam		Dual Beams		
Detection distance	Outdoor	20m	40m	60m	100m	100m
	Indoor	60m	120m	180m	300m	300m
Detection distance(max)		120m	250m	350m	600m	600m
Detection method		Simultaneous interruption of infrared beams				
Interruption time		40ms~1000ms(adjustable)				
Frequencies		3different frequencies (selectable)				
Power and voltage		DC9-30V/AC9-20V				
Current consumption		70mA max				
Alarm cycle		≥1.5s				
Alarm output		NO / NC relay outputs				
IP rating		IP65				
Operating temperature		-25℃~55℃				
Humidity		95% max				
Correction angle		Horizontal 180°(±90°), Vertical 20°(±10°)				
Install location		Indoor/Outdoor ,Wall/Pole				
Weight		650g		950g		
Attachment	Wall mounting screw	4pcs , PM4*25mm				
	Expansion pipe	4pcs, Φ7*27mm, green				

12.Dimensions

AB0-i series



ABT-i series

